Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Linear Patterns

Name _____ Date _____ No calculator Version 1

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KEY WORDS pattern *(tauira)* · sequence *(raupapa)* · rule *(ture)* · term *(tūnga)*

A Continuing a pattern ●○○

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 3, 7, 11, 15, ___, ___

b) 5, 11, 17, 23, ___, ___

c) 20, 17, 14, 11, ___, ___

d) 1, 3.5, 6, 8.5, ___, ___

e) −10, −6, −2, 2, ___, ___

f) 50, 43, 36, 29, ___, ___

g) 0.2, 0.5, 0.8, 1.1, ___, ___

h) 8, 19, 30, 41, ___, ___

i) 100, 85, 70, 55, ___, ___

j) −1, −3, −5, −7, ___, ___

B The constant change ●○○

2. What is added or subtracted each time?

a) 4, 7, 10, 13, 16

b) 10, 18, 26, 34, 42

c) 30, 25, 20, 15, 10

d) 7, 7.5, 8, 8.5, 9

e) −6, −4, −2, 0, 2

f) 45, 36, 27, 18, 9

g) 2, 15, 28, 41, 54

h) 1.5, 1, 0.5, 0, −0.5

C Using a rule ●●○

Remember: In the rule $T = 3n + 2$, n is the position of the term. For the 4th term, put $n = 4$: $3 \times 4 + 2 = 14$.

3. Write the first four terms of the pattern with each rule.

a) $T = 3n + 2$

b) $T = 5n - 1$

c) $T = 2n + 7$

d) $T = 4n$

e) $T = 10n - 6$

f) $T = -2n + 20$

4. Find the term asked for.

a) $T = 3n + 2$, the 10th term

b) $T = 5n - 1$, the 20th term

c) $T = 4n + 7$, the 50th term

d) $T = -2n + 100$, the 30th term

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D Finding the rule ●●●

> **Remember:** The number in front of n is the constant change. Then work out what must be added or subtracted to make the first term correct.

5. Find the rule for the nth term of each pattern.

a) 5, 8, 11, 14, 17

b) 7, 9, 11, 13, 15

c) 4, 8, 12, 16, 20

d) 1, 6, 11, 16, 21

e) 10, 16, 22, 28, 34

f) 3, 13, 23, 33, 43

g) 20, 18, 16, 14, 12

h) 15, 11, 7, 3, −1

E Patterns in context ●●●

6. Squares are made in a row from matchsticks. The table shows the number of matchsticks in each pattern.

Pattern	1	2	3	4
Matchsticks	4	7	10	13

a) Find the number for pattern 5.

b) Write the rule for pattern n.

c) Find the number for pattern 20.

d) Which pattern has 46 matchsticks?

7. Tables are joined in a line for a party. The table shows how many people can sit at each number of tables.

Tables	1	2	3	4
People	6	10	14	18

a) Find the number for tables 5.

b) Write the rule for tables n.

c) Find the number for tables 12.

d) Which tables has 38 people?

★ Extension

8. A pattern has the rule $T = 4n + 2$. Is 100 a term of this pattern? Show your working.

9. Tama has \$50 in the bank and saves \$15 every week. After n weeks he has $15n + 50$ dollars. After how many weeks will he have \$200?

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1-1 PRIVATE TUTORS

ALGEBRA

Linear Patterns

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS pattern *(tauira)* · sequence *(raupapa)* · rule *(ture)* · term *(tūnga)*

A Continuing a pattern ●○○

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 2, 7, 12, 17, ___, ___

b) 4, 11, 18, 25, ___, ___

c) 25, 21, 17, 13, ___, ___

d) 2, 3.5, 5, 6.5, ___, ___

e) −12, −7, −2, 3, ___, ___

f) 60, 51, 42, 33, ___, ___

g) 0.1, 0.5, 0.9, 1.3, ___, ___

h) 7, 19, 31, 43, ___, ___

i) 90, 75, 60, 45, ___, ___

j) −2, −5, −8, −11, ___, ___

B The constant change ●○○

2. What is added or subtracted each time?

a) 5, 9, 13, 17, 21

b) 12, 21, 30, 39, 48

c) 40, 34, 28, 22, 16

d) 3, 3.5, 4, 4.5, 5

e) −8, −5, −2, 1, 4

f) 54, 46, 38, 30, 22

g) 3, 17, 31, 45, 59

h) 2.5, 2, 1.5, 1, 0.5

C Using a rule ●●○

Remember: In the rule $T = 3n + 2$, n is the position of the term. For the 4th term, put $n = 4$: $3 \times 4 + 2 = 14$.

3. Write the first four terms of the pattern with each rule.

a) $T = 4n + 1$

b) $T = 3n - 2$

c) $T = 2n + 9$

d) $T = 6n$

e) $T = 10n - 4$

f) $T = -3n + 30$

4. Find the term asked for.

a) $T = 4n + 1$, the 10th term

b) $T = 3n - 2$, the 20th term

c) $T = 6n + 5$, the 50th term

d) $T = -3n + 200$, the 40th term

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D Finding the rule ●●●

> **Remember:** The number in front of n is the constant change. Then work out what must be added or subtracted to make the first term correct.

5. Find the rule for the nth term of each pattern.

a) 6, 10, 14, 18, 22

b) 5, 7, 9, 11, 13

c) 3, 6, 9, 12, 15

d) 2, 7, 12, 17, 22

e) 9, 16, 23, 30, 37

f) 4, 14, 24, 34, 44

g) 30, 27, 24, 21, 18

h) 18, 13, 8, 3, −2

E Patterns in context ●●●

6. Triangles are made in a row from matchsticks. The table shows the number of matchsticks in each pattern.

Pattern	1	2	3	4
Matchsticks	3	5	7	9

a) Find the number for pattern 5.

b) Write the rule for pattern n.

c) Find the number for pattern 25.

d) Which pattern has 37 matchsticks?

7. Tables are joined in a line for a party. The table shows how many people can sit at each number of tables.

Tables	1	2	3	4
People	8	14	20	26

a) Find the number for tables 5.

b) Write the rule for tables n.

c) Find the number for tables 10.

d) Which tables has 44 people?

★ Extension

8. A pattern has the rule $T = 5n + 3$. Is 120 a term of this pattern? Show your working.

9. Tama has \$40 in the bank and saves \$20 every week. After n weeks he has $20n + 40$ dollars. After how many weeks will he have \$300?

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